Sahaj Bhadja

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Summary

Software Engineer with 2+ years of experience building **TypeScript/Node products**, **payment APIs**, and reliable **customer-facing platforms** with modern CI/CD workflows.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Web Technologies, Analysis of Algorithms

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Object-Oriented Programming, Operating Systems

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Reduced dashboard latency from 1.2s to 350ms by optimizing indexed queries and **multi-tenant data models** for an Azure SQL platform supporting 500+ users with row-level isolation controls.
- Achieved 99%+ reliability across 5K+ monthly jobs by building **fault-tolerant FastAPI pipelines** with validation, reconciliation, error isolation, cloud storage, incremental processing, and stable repeatable CI/CD-backed production releases.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Reduced feedback turnaround by 60% through a **customer-facing Python platform** supporting 150+ instructors and 10K+ daily requests with REST APIs, AWS RDS, secure ingestion, and CI/CD workflows.
- Processed 1K+ files per batch using **Python multithreading**, validating PDF, DOCX, and ZIP inputs while isolating failures and preserving responsive product workflows under concurrent production load reliably.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) *India*

- Improved API response times by 30% through **maintainable microservices**, indexing, caching, pagination, and filtered payloads while troubleshooting integrations across Windows, Linux, and macOS production environments at scale.
- Increased pipeline efficiency by 68% using **AWS Glue and S3**, automating builds and 20+ unit/integration suites through Jenkins, PyTest, Mockito, GitHub, reporting, and failure diagnostics at scale.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) *India*

- Synchronized 2M+ records through an eight-service **Node.js microservice platform** covering authentication, customers, billing, payments, resources, leads, permissions, and notifications with WebSockets and Azure SQL across distributed locations.
- Reduced resource wastage by 42% through **real-time product workflows**, implementing JWT controls, React interfaces, Azure monitoring, automated tests, and Docker/Jenkins deployments across modular service domains and production releases.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Delivered sub-200ms **Flask payment APIs** across 50K+ booking records, implementing pricing, coupons, Firebase transactions, and real-time status synchronization for TypeScript-backed mobile and web customer experiences at scale.
- Reduced manual refund handling by 70% through **payment webhook automation**, integrating cards, wallets, disputes, JWT access, and hourly Databricks/Spark reporting pipelines while aligning architecture with customer requirements.

Skills

Languages: TypeScript, JavaScript, Node.js, Python, Java, SQL, C++
Product & Backend: React, Next.js, Flask, FastAPI, REST APIs, Microservices, WebSockets, System Design, Data Modeling
Payments & Security: Payment APIs, Webhooks, Refund Workflows, Dispute Handling, Plaid, JWT, RBAC, Data Validation
Cloud & Data: AWS, Azure, PostgreSQL, Azure SQL, AWS RDS, AWS S3, MongoDB, Firebase
Engineering: CI/CD, GitHub Actions, Jenkins, Git, GitHub, Docker, Kubernetes, PyTest, Mockito, Unit Testing, Integration Testing, Troubleshooting

Projects

Payment Lifecycle Orchestrator - Design Concept

- Proposed a **TypeScript/Node payment service** modeling authorization, settlement, refunds, and chargebacks with signed webhooks, idempotency keys, explicit state transitions, retry policies, audit logs, and merchant-facing status APIs.

Merchant Reconciliation Portal - Design Concept

- Proposed a **customer-facing reporting platform** reconciling payment events against settlements, surfacing discrepancies, searchable transaction histories, export workflows, role-based access, and operational dashboards through React and PostgreSQL-backed APIs.

FinNexus | Live

- Built a **TypeScript fintech platform** supporting 50K+ users and 1M+ transactions with Plaid bank linking, token-based security, MongoDB, WebSockets, real-time synchronization, filtering, Kubernetes, and CI/CD delivery workflows.